

Komaba campus, the University of Tokyo

Revealing the inner structure of Japanese echo-words:

An idiosyncratic form of reduplication



Mamoru Yakuwa¹, Izumi Matsumiya²

¹Aoyama Gakuin University, ²Japan Women's University

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Roadmap

- 1 Introduction
- 2 Study 1: Segmental distribution replicated
- 3 Liaison of studies
- 4 Study 2: Alternation and representational effect
- 5 Discussion
- 6 Conclusion

Echo words: An empirical introduction

- **Echo-word**: a morphological phenomenon involving incomplete reduplication.
- Although echo-words have not been fully described, they are observed cross-linguistically.

[base] → reduplication → [base][reduplicant] / [reduplicant][base]

(1) *Reduplication Meduplication* in Khalkha-Mongolian Kubo (1997)

- | | | | |
|------------------|---|--------------------|-----------------------------|
| a. Japon 'Japan' | → | Japon Mapon | 'Japan and other countries' |
| b. aav 'father' | → | aav maav | 'father and other people' |
| c. mašin 'car' | → | mašin zašin | 'car(s) and other things' |
| d. max 'meat' | → | max zax | 'meat and other things' |

Echo-words in Japanese

- Imperfect reduplication involving a change in the consonant and/or vowel of the initial syllable of the second member ([base] + [reduplicant]).

(2) Echo-reduplication

- Øuya-**m**uya 'frustrated'
- k**ati-**k**oti 'frozen'
- m**aze-**k**oze 'mixed'

Note

- Perfect reduplication (e.g., *doki-doki* and *pati-pati*) and dvandva words (e.g., *are-kore* and *siro-kuro*), which consist of two or more words that can be connected by 'and', are excluded from this study.

Previous studies

Descriptive studies: Murata (1984), Labrune (2006), (Kwon & Masuda, 2019)

Segmental strength: **base** $\leq$ **reduplicant**

Sound hierarchy (Murata, 1984)

- He defines a hierarchy of segmental strength *based on echo-word alternations*.
- Unlike the *sonority hierarchy*, this hierarchy is taken to reflect the *psychological reality* of native speakers.

Previous studies

(4) 母音ハイレラーキ :

i < u < e < a < o

(5) 子音ハイレラーキ :

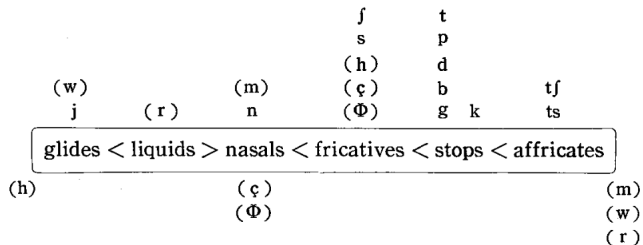


Fig.1 Sound hierarchy in Japanese (Murata, 1984, 70).

The segment in parentheses indicates its position in the sonority hierarchy.

(3) Examples observed in the experiment by Murata (1984)

a. *zuu-zaa* > *zaa-zuu*

b. *yuura-suura* > *suura-yuura*

Previous studies

Phonemic preferences: “Weak” consonants occur in N1-initial position, and “strong” ones in N2-initial position.

- (4) Phonemic preferences in Japanese echo-words (Labrune, 2006)¹
Frequencies decrease from left to right.

- a. Initial segment in the first half (C1)

k p g h s b z r y w d n ' m t

- b. Initial segment in the last half (C2)

g ' z n d t p y w b r m h s k

- (5) Example

- a. ***maze-koze*** > ***koze-maze***

Previous studies

- Segmental strength
 - ① **Segments lower in sonority**: e.g., obstruents /t/, /k/, /s/, etc.
 - ② **Segments requiring more articulatory effort**: e.g., /k/, /h/, /r/, etc.
- Similar tendencies are found in abbreviated loanword compounds (*poke-mon* ‘pocket monster’) and dvandvas (*are-kore* ‘this and that’).

¹ Boldface indicates statistically significant segments. (χ^2 -test)

Research questions and our aims

Problem: Previous studies describe the patterns of echo-words well; however, *they do not explain why these patterns arise or what criteria underlie them.*

Research question: *What criterion determines the “weak–strong” pattern?*

● Our aims

- Provide a modern update of Labrune (2006)’s account.
- Identify the criterion for segmental strength.
- Provide a theoretical explanation.

Method

Our data

- The data were drawn from 『擬音語・擬態語 4500 日本語オノマトペ 辞典』 (Ono, 2007).
- *Ideophonic words that exhibit imperfect reduplication through a change in the consonant and/or vowel of the initial syllable of the second member* (Labrune, 2006).
- $\neq$ dvandva words
- $N = 130$



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Raw data

Frequencies of consonants and vowels

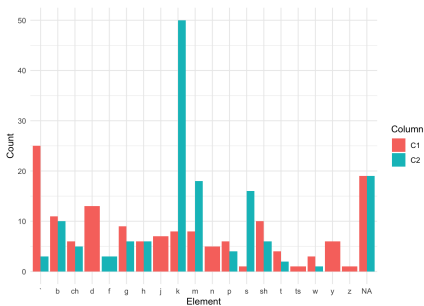


Fig.2 C1 and C2 frequencies

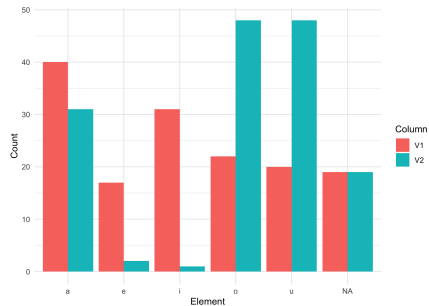


Fig.3 V1 and V2 frequencies

Inferential analysis

Bayesian Poisson mixed effects regression model

- Motivation for using this model.
 - Bayesian inference is appropriate for a relatively small sample size.
 - Poisson regression is appropriate for **count data** (see Winter & Bürkner, 2021).
 - The model provides a methodological update of Labrune (2006).
- Model details.
 - **Implementation:** `brms` package (Bürkner, 2017) on R (R Core Team, 2025).
 - **Prior:** $b \sim \mathcal{N}(\mu = 0, \sigma = 1)$
In prose: This is a weakly informative prior for the coefficients (Lemoine, 2019).
 - **MCMC:** `iter = 2000*4 chains, warmup = 1000, thin = 1`

Inferential analysis

Model formula and parameters

Formula: $\text{freq} \sim \text{segment} * \text{pos} + \text{offset}(\log(\text{phon_freq}))$

Parameters

- **dependent variable**
 - *freq*: Frequency in each position (C1/C2, V1/V2)
- **independent variables**
 - *segment*: Each consonantal/vocalic segment
 - *pos*: Position
 - *segment * pos*: Interaction between the two variables
- **Offset variable** (see Winter & Bürkner, 2021, Sect. 3.4)
 - $\log(\text{phon_freq})$: Controls for the effect of phonemic frequency in Japanese

Results: consonants

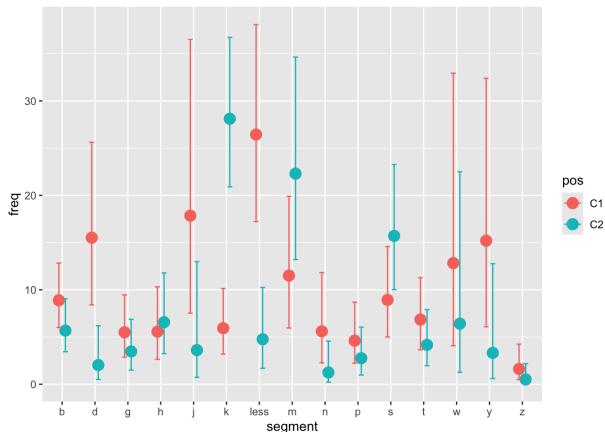


Fig.4 Posterior probability of frequency for each consonant

Tab.1 Regression results for significant C1 parameters whose 95% CIs did not include 0.

predictor	β	Error	95% CI	pd
Intercept	-3.42	0.19	[-3.81 – -3.05]	100%
C1: less	1.08	0.27	[0.56 – 1.59]	99.95%
C1: z	-1.73	0.57	[-2.90 – -0.68]	99.98%

Tab.2 Regression results for significant C2 parameters whose 95% CIs did not include 0.

predictor	β	Error	95% CI	pd
C2: d	-1.61	0.66	[-2.96 – -0.39]	99.62%
C2: k	2.01	0.38	[1.26 – 2.79]	100%
C2: less	-1.29	0.52	[-2.33 – -0.33]	100%
C2: m	1.11	0.43	[0.29 – 1.96]	99.52%
C2: s	1.02	0.40	[0.23 – 1.81]	99.48%

Results: vowels

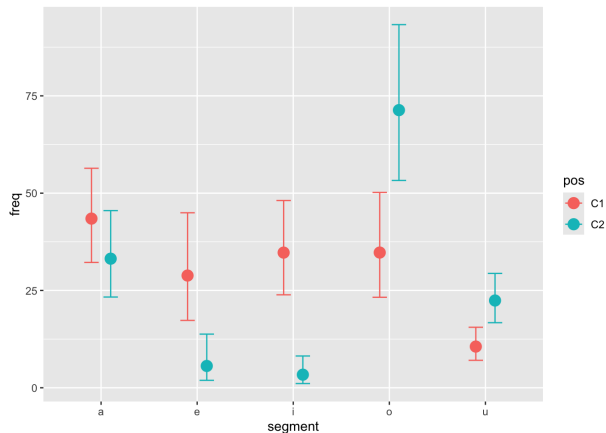


Fig.5 Posterior probability of frequency for each vowel

Tab.3 Regression results for significant V1 parameters whose 95% CIs did not include 0.

predictor	β	Error	95% CI	pd
Intercept	-0.06	0.14	[-0.35 – 0.21]	65%
V1: u	-1.41	0.24	[-1.89 – -0.93]	100%

Tab.4 Regression results for significant V2 parameters whose 95% CIs did not include 0.

predictor	β	Error	95% CI	pd
V2: e	-1.38	0.55	[-2.49 – -0.33]	99.55%
V2: i	-2.09	0.55	[-3.22 – -1.07]	100%
V2: o	0.99	0.31	[0.38 – 1.59]	99.98%
V2: u	1.02	0.32	[0.40 – 1.67]	99.92%

Interim summary

Consonants

- Largely consistent with the literature.
- Significantly less frequent segments

C1: z, d; C2: d, ∅

- Significantly frequent segments

C1: ∅; C2: k, m, s

Vowels

- Significantly less frequent segments

C1: u; C2: e, i

- Significantly frequent segments

C2: o, u

Representation of segments

Element Theory (Backley, 2011)

- *Monovalency*: the presence or absence of a property
- *Combinability*: represented as a single element or in combination
- *six elements*: vowel set { |I, U, A| }, consonant set { |H, N, ?| }

Tab.5 Characteristics of elements based on Backley and Nasukawa (2010)

Element	in onset	in nucleus
I	palatal, coronal POA	front vowels
U	labial, velar POA	rounded vowels
A	uvular, pharyngeal POA	non-high vowels
H	high tone	aspiration, voicelessness
N	nasality, low tone	nasality, obstruent voicing
?	creaky voice (laryngealised vowels)	oral or glottal occlusion

Element representation

- Configuration based on Backley and Nasukawa (2010).
- Headedness is *not considered*.

Tab.6 Vowels

Vowel	Element
a	A
e	A I
i	
o	A U
u	U

Tab.7 Consonants

Consonant	Element	Consonant	Element
k	H ? U	p	H ? U
g	N ? U	b	N ? U
s	H ? I	m	N U
z	N ? I	r	N ? I
t	H ? I	w	U
d	N ? I	y	
n	N I	f	H ? U
h	H ? I	ç	H ? I
ʃ	H ? I	ʒ	N ? I

Representational effect

Five types of element alternation

Tab.8 Our classification

Type	Vowel example	Consonant example
Addition	<i>gasa-goso</i> / A → A U	front vowels
Deletion	<i>dogi-magi</i> A U → A	rounded vowels
Full replacement	<i>ata-huta</i> A → U	non-high vowels
Partial replacement	<i>deko-boko</i> A I → A U	voicelessness
No change	<i>musha-kusha</i> U → U	nasality, obstruent voicing

Motivations for adopting elements rather than distinctive features

- Representations can be illustrated with a simple *presence/absence contrast*.
- The system uses combinations of only *six elements*.

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Raw data

Frequencies of consonants and vowels

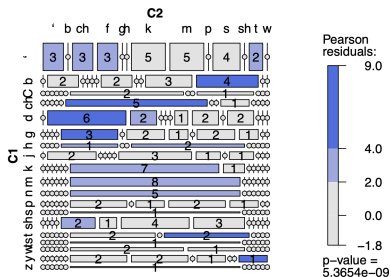


Fig.6 Mosaic plot of C1 - C2 alternation

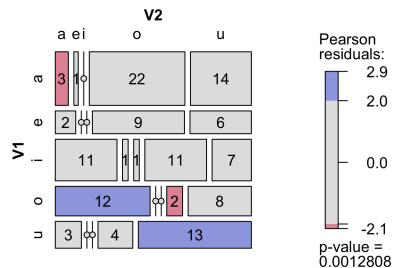


Fig.7 Mosaic plot of V1 - V2 alternation

Raw data

Types of element alternation

Tab.9 Alternation patterns observed for vowels

Pattern	Count
Addition	29
Deletion	22
Full replacement	50
Partial replacement	9
No change (Consonant change)	20
Total	130

Tab.10 Alternation patterns observed for consonants

Pattern	Count
Addition	24
Deletion	1
Full replacement	15
Partial replacement	71
No change (Consonant change)	19
Total	130

Inferential analysis

Bayesian cumulative mixed effects logistic regression model

- Aims
 - The phonetic hierarchy is represented as points (0–8 for C, 1–5 for V).
 - Bayesian CLMM for an ordinal dependent variable.
- Model details
 - **Implementation:** `brms` (Bürkner, 2017) on R (R Core Team, 2025).
 - **Priors:**
 - ① $b \sim \mathcal{N}(0, 1)$: Weakly informative prior for coefficients
 - ② Intercept $\sim t_3(0, 2)$: Student's t prior for fixed intercepts
 - ③ SD $\sim t_3(0, 1)$: Student's t prior for random intercepts
 - **MCMC:** `iter = 4000 * 4 chains`, `warmup = 2000`, `thin = 1`
 - Sum-to-zero coding using the `contrast()` function

Inferential analysis

Model formula and parameters

Formula: $point2 \sim scale(point1) + scale(Def_point) + el_no * el_pat + (1 | C1/V1) + (1 | Type)$

Parameters

- **dependent variable**

- *point2*: Hierarchy point for C2 (0–8)/V2 (1–5)

- **independent variables**

- *scale(point1)*: Standardized point for C1 (0–8)/V1 (1–5)
- *scale(Def_point)*: Standardized difference between *point1* and *point2*
- *el_no*: Number of elements added or deleted
- *el_pat*: Five patterns of element alternation

- **Random intercepts**

- $(1 | C1/V1)$: Each segment
- $(1 | Type)$: Three types of echo-word

MCMC summary

The two models converged properly, with $\hat{R} = 1.00$ and robust ESS values for all parameters.

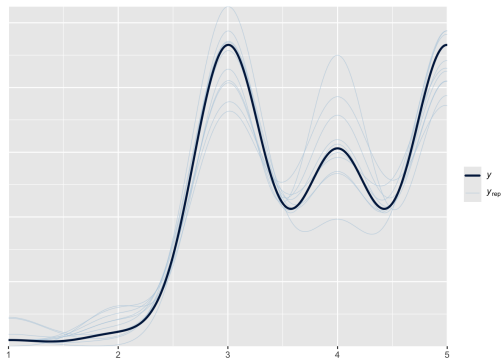


Fig.8 Four-chain MCMC sampling for the vowel model

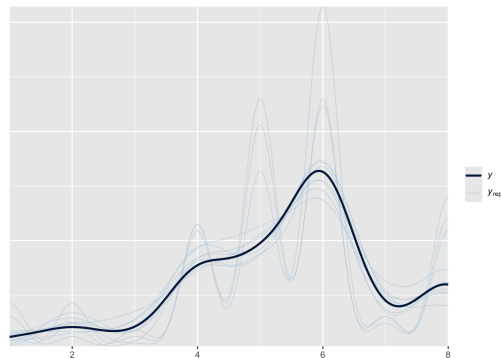


Fig.9 Four-chain MCMC sampling for the consonant model

Results: vowels

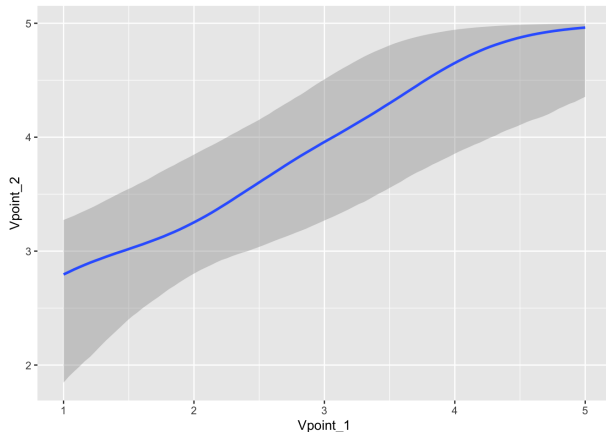


Fig.10 Predicted effect and 95% CI of the V1 point on the V2 point

Tab.11 Relationship between the two vowel points

predictor	β	Error	95% CI	pd
scale(Vpoint_1)	3.64	0.68	[2.10 – 4.82]	100%

- There is a positive and robust correlation between the two vowels.
- The stronger V1 is, the stronger V2 tends to be.
- Note: V2 is inherently stronger than V1.

Results: vowels

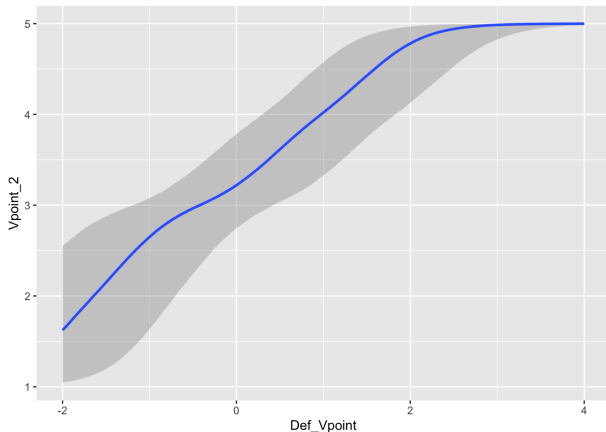


Fig.11 Predicted effect and 95% CI of point difference on the V2 point

Tab.12 Effect of point difference

predictor	β	Error	95% CI	pd
scale(DefVpoint_1)	5.26	0.57	[4.17 – 6.40]	100%

- There is a positive and robust correlation between the V2 point and the point difference.
- The stronger V2 is, the larger the gap between the two vowels becomes.

Results: vowels

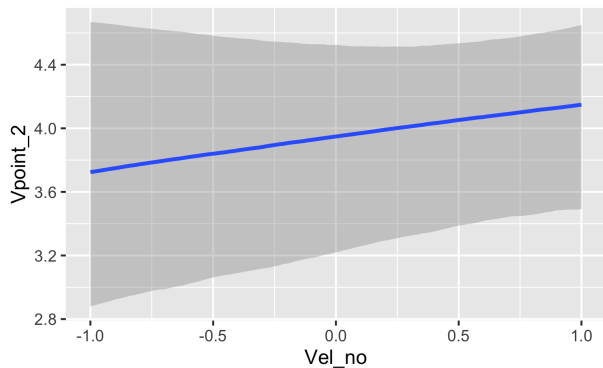


Fig.12 Predicted effect and 95% CI of the number of added elements on the V2 point

Tab.13 Effect of element quantity on the V2 point

predictor	β	Error	95% CI	pd
scale(Vel_no)	0.70	0.70	[-0.63 – 2.12]	71.45%

- Uncertain effect: the magnitude of β is small, and the CI includes 0.
- There is a slightly positive effect for an increasing number of elements.
- As the number of elements increases from V1 to V2, the V2 point tends to become higher, though this effect is uncertain.

Results: vowels

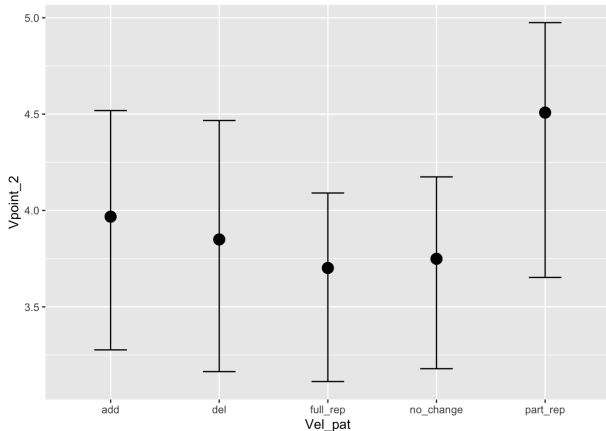


Fig.13 Mean and 95% CI of the posterior probability for each element alternation pattern

Tab.14 Effect of element quantity on the V2 point

predictor	β	Error	95% CI	pd
add	0.08	0.85	[-1.59 – 1.77]	53.57%
del	-0.16	0.84	[-1.81 – 1.50]	57.88%
full_rep	0.78	0.76	[-0.70 – 2.26]	84.71%
no_change	0.00	0.98	[-1.93 – 1.90]	50.01%

Tab.15 Bayesian hypothesis tests among element-pattern parameters

Hypothesis	β	Error	90% CI	Post. Prob.
part_rep - full_rep > 0	2.55	1.73	[-0.03 – 5.54]	95%(*)

- Partial replacement is almost significantly probable than full replacement.

Results: consonants

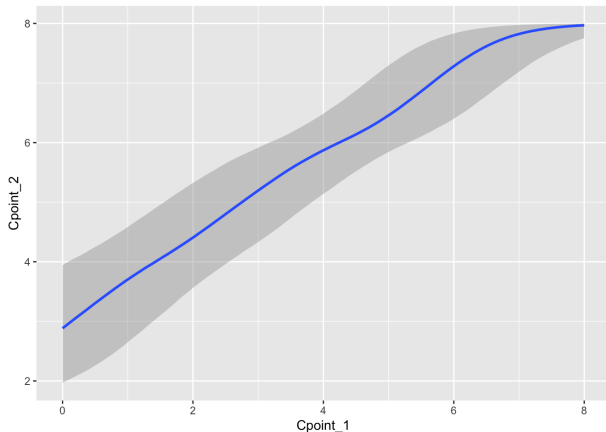


Fig.14 Predicted effect and 95% CI of the C1 point on the C2 point

Tab.16 Relationship between the two consonant points

predictor	β	Error	95% CI	pd
scale(Cpoint_1)	4.96	0.48	[4.02 – 5.91]	100%

- There is a positive and robust correlation between the two consonants.
- The stronger C1 is, the stronger C2 tends to be.
- Note: C2 is inherently stronger than C1.

Results: consonants

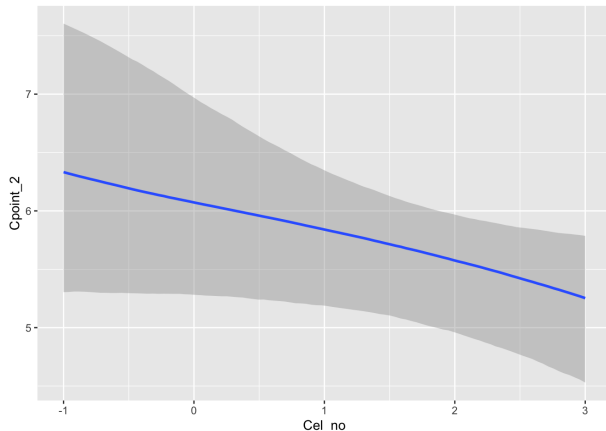


Fig.16 Predicted effect and 95% CI of the number of deleted elements on the C2 point

Tab.18 Effect of element quantity on the C2 point

predictor	β	Error	95% CI	pd
scale(Cel_no)	-0.52	0.46	[-1.44 – 0.35]	86.58%

- Uncertain effect: the magnitude of β is small, and the CI includes 0.
- There is a slightly positive effect for a decreasing number of elements.
- As the number of elements decreases from C1 to C2, the C2 point tends to become higher, though this effect is uncertain.

Results: consonants

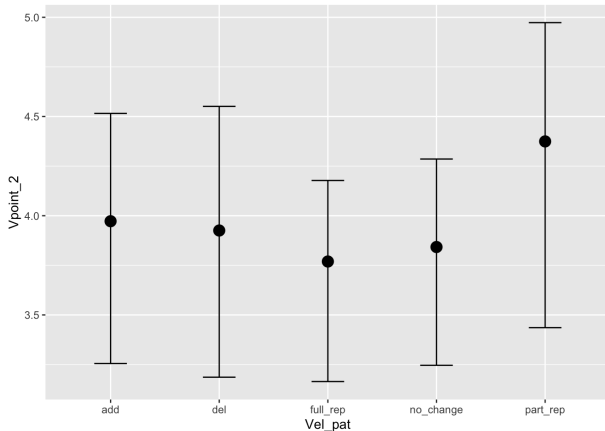


Fig.17 Mean and 95% CI of the posterior probability for each element alternation pattern

Tab.19 Effect of element quantity on the C2 point

predictor	β	Error	95% CI	pd
add	0.71	0.81	[-0.92 – 2.31]	79.94%
del	0.30	0.88	[-1.41 – 2.08]	62.68%
full_rep	0.69	0.61	[-0.50 – 1.90]	86.98%
no_change	-0.90	0.55	[-2.01 – 0.19]	94.47%

Tab.20 Bayesian hypothesis tests among element-pattern parameters

Hypothesis	β	Error	90% CI	Post. Prob.
part_rep - full_rep > 0	1.49	0.86	[0.11 – 2.96]	96%*

- In contrast to vowels, full replacement is favored.

Recap from Study 2

- ① *Positive correlations between the V1–V2 and C1–C2 points.*
- ② *Positive correlations between point values and point differences.*
- ③ *Opposite effects of the number of elements*
 - A weakly positive correlation between point values and the number of elements *for vowels*.
 - A weakly negative correlation *for consonants*.

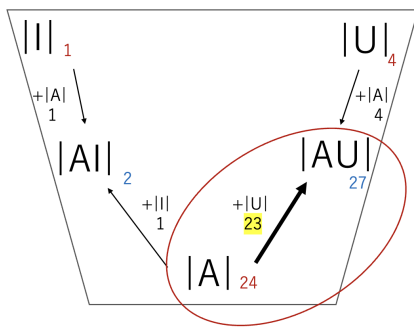
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Element alternations (Vowel only)

The |U| element behaves differently from the other elements in echo-word formation.

Addition



- The number of |U| elements increases from V1 to V2.
- *The idiosyncrasy of the |U| element.*

Fig.18 Element alternation under full replacement.

Element alternations (Vowel only)

Partial Replacement

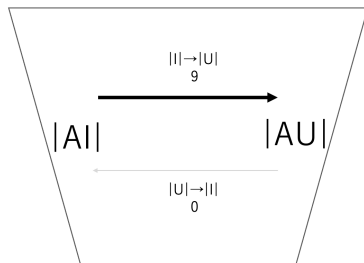


Fig.19 Element alternation under addition. Red numbers refer to V1 and blue numbers to V2.

Full Replacement

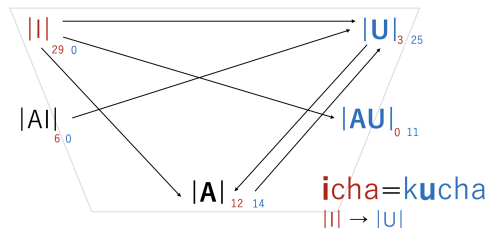


Fig.20 Element alternation under partial replacement.

Idiosyncrasy of the |U| element

Where does the idiosyncrasy of |U| come from?

Idiosyncrasy of |U| in Japanese?

- The activity of |U| is *naturally weak* in (Tokyo) Japanese (Buckley & Nasukawa, 2016). → Is it a marked element in Japanese?

This weakness of |U| in Japanese may be related to our results.

Conclusion

- 1 We successfully replicated the *C1–C2* preferences discussed in Labrune (2006) using a different method.
- 2 The strength of *C1/V1* is *positively correlated* with that of *C2/V2*.
- 3 As *C2/V2* becomes stronger, the strength gap from *C1/V1* becomes wider.
- 4 As *C2/V2* becomes stronger, the number of elements *slightly increases in vowels* and *decreases in consonants*. (This result is still statistically uncertain.)
- 5 The behavior of the *|U|* element is *different from that of the other elements*.

Residual issues and future challenges

Deletion

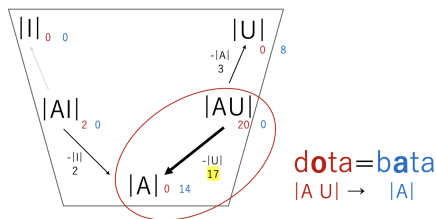


Fig.21 Element alternation under deletion.

- In element deletion, the $|U|$ element tends to be deleted most often.
- How can it be explained?

Residual issues and future challenges

Future challenges

- **Typological perspective:**

Languages whose echo patterns are fixed (e.g., Khalkha Mongolian) vs. languages whose patterns are arbitrary or complex (e.g., Japanese and Hungarian (Patay, 2017, 2018))

- **Relevance to phonetic detail:**

Element theory can represent phonetic or acoustic detail.

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